CASE

Casey, 19 · 12 hours of fever, headache, neck stiffness, photophobia · purple rash on legs · positive Kernig and Brudzinski signs · LP: cloudy CSF, ↑↑ WBC (neutrophils), ↑ protein, ↓ glucose · **bacterial meningitis**

BIO 004 · WEEK 7 · CASE DEEP DIVE

CNS Meninges and CSF

A college student with a fever, a stiff neck, and a small purple rash. A lumbar puncture in the back. A bacterial infection in the fluid that bathes the brain.

PATIENT CASE

Casey is 19, a college freshman. Twelve hours ago she developed a **severe headache, fever, and neck stiffness**. Her roommate brought her to the ED when she became increasingly drowsy.

On exam she is febrile (39.2°C), photophobic, with **resistance to neck flexion**. She has a **petechial rash on her lower legs**. Kernig's and Brudzinski's signs are positive. After a quick head CT to rule out mass effect, the team performs a **lumbar puncture**. CSF is **cloudy** with **2,500 WBC (90% neutrophils), elevated protein, glucose 25 mg/dL (low)**. Gram stain shows gram-negative diplococci. The diagnosis is **bacterial meningitis, likely Neisseria meningitidis**. Empiric IV antibiotics started before she leaves the procedure room.

Casey's case rests on the layers wrapping the central nervous system, the space between them, and the fluid that flows through it.

> GUIDING QUESTION
>
> What are the three meningeal layers, where does CSF live, and what does a Gram stain in CSF tell you that you couldn't learn any other way?

PREWORK Companion to your Week 7 CNS Meninges & CSF prework page.

1 THREE MENINGES

Layered protection from outside in

Dura mater (outer)

Tough fibrous "tough mother." Two layers in the cranium (periosteal + meningeal); the outer periosteal layer adheres to the inside of the skull. Forms folds (falx cerebri between hemispheres, tentorium cerebelli over cerebellum) that compartmentalize the brain.

Arachnoid mater (middle)

Spider-web-like layer below the dura. Web-like trabeculae cross the subarachnoid space.

Pia mater (inner)

Thin "tender mother." Adheres to brain and spinal cord surface, following every gyrus and sulcus.

Spaces between layers:

Epidural space

Between skull/vertebrae and dura. Potential space; contains middle meningeal artery branches. Site of epidural hematoma (arterial bleed, often after temporal bone fracture).

Subdural space

Between dura and arachnoid. Bridging veins cross here. Tearing them produces subdural hematoma (venous bleed, slow, common in elderly after falls).

Subarachnoid space

Between arachnoid and pia. Filled with CSF. Contains cerebral arteries (rupture = subarachnoid hemorrhage). Where bacteria sit in meningitis.

2 CSF PRODUCTION

Made by choroid plexus, ~500 mL/day

Cerebrospinal fluid is produced by the **choroid plexus**, a specialized vascular structure hanging into each ventricle. Ependymal cells covering capillary tufts actively transport fluid and ions from blood into the ventricles. Total CSF volume at any time is ~125-150 mL; production ~500 mL/day. The fluid is fully turned over 3-4 times daily.

CSF composition: similar to plasma but with much lower protein (15-45 mg/dL vs ~7,000 in plasma) and lower glucose (~2/3 of serum glucose). Almost no cells (<5 lymphocytes per mm^3). Crystal clear.

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3 CSF FLOW

From ventricles to subarachnoid to venous blood

1. Produced in **lateral ventricles** (one per hemisphere).
2. Flows through **interventricular foramina (of Monro)** into the **third ventricle**.
3. Through the narrow **cerebral aqueduct (of Sylvius)** into the **fourth ventricle**.
4. Exits the 4th ventricle through three openings (two lateral foramina of Luschka, one median foramen of Magendie) into the **subarachnoid space**.
5. Circulates around brain and spinal cord (down the central canal too).
6. Reabsorbed into venous blood via **arachnoid granulations** projecting into the superior sagittal sinus.

Obstruction anywhere → **hydrocephalus** (CSF accumulation, raised intracranial pressure).

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4 BLOOD-BRAIN BARRIER

The neighborhood's tight border

The brain's capillaries are unique: **continuous capillaries with tight junctions, surrounded by astrocyte end-feet**. The combination creates the *blood-brain barrier*, which restricts what crosses from blood into brain tissue. Small lipid-soluble molecules (O2, CO2, alcohol, anesthetics) cross easily. Most other substances need specific transporters or can't cross at all. The BBB is what protects the brain from most circulating toxins and pathogens — and what makes treating CNS infections and tumors so difficult (drugs that don't cross are useless).

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5 LUMBAR PUNCTURE

How to sample CSF safely

The spinal cord ends at ~L1-L2 in adults. Below that, the *cauda equina* (lumbar and sacral nerve roots) floats freely in CSF. A needle inserted between L3-L4 or L4-L5 (just below the iliac crest) passes through skin, subcutaneous tissue, supraspinous and interspinous ligaments, ligamentum flavum, dura, and arachnoid — into the subarachnoid space. No cord is at risk (the nerve roots get pushed aside).

CSF dripping out is sent for: cell count and differential, protein, glucose, Gram stain, culture, sometimes PCR for specific pathogens.

What CSF looks like in different infections

TYPE	WBC	PREDOMINANT CELL	PROTEIN	GLUCOSE
Normal	<5/mm³	lymphocytes	15-45 mg/dL	~2/3 serum
Bacterial (Casey)	1,000-10,000	neutrophils	↑↑	↓↓ (bacteria consume it)
Viral	10-1,000	lymphocytes	normal/mild ↑	normal
Fungal/TB	50-500	lymphocytes	↑	↓

Why the rash matters. Petechial/purpuric rash with meningitis = Neisseria meningitidis until proven otherwise. The bacterium produces endotoxin causing disseminated intravascular coagulation. Patients can deteriorate from awake to dead in hours. Antibiotics should NOT wait for LP results.

✎ **DRAW ZONE 1** *-- Sketch a coronal section through the head showing the three meninges, the three potential spaces, and a vessel in each.*

Color each layer. Mark where epidural (arterial), subdural (venous), and subarachnoid (CSF + bacteria) bleeds/infections live.

DRAW ZONE 2

✎ -- *Diagram CSF flow from production to reabsorption. Number the steps from choroid plexus through ventricles, foramina, and back into venous blood via arachnoid granulations.*

Star common sites of obstruction (cerebral aqueduct narrow = hydrocephalus).

RETURNING TO CASEY

An infection in the fluid that protects the brain

Casey's CSF profile is bacterial meningitis with a particularly dangerous organism. Every classical sign she has maps to inflammation of the meninges and the subarachnoid space.

"*Severe headache + photophobia*"

Inflamed meninges are exquisitely pain-sensitive. Light worsens pain because pupillary response engages inflamed structures.

"*Neck stiffness, Kernig, Brudzinski*"

Stretching inflamed meninges causes pain. Patients reflexively guard against it. Specific physical signs for meningeal irritation.

"*Cloudy CSF with 90% neutrophils*"

Neutrophilic response to bacterial infection. Cloudiness is the high cell count plus protein plus organisms making the fluid opaque.

"*Low CSF glucose*"

Bacteria consume glucose. Glucose <40% of serum value is highly suggestive of bacterial (or sometimes fungal/TB) meningitis.

"*Petechial rash*"

Neisseria meningitidis endotoxin triggers DIC. Petechiae are tiny hemorrhages from consumed clotting factors. A red flag for catastrophic meningococcal disease.

"*Why antibiotics before LP results*"

Bacterial meningitis can kill within hours. Empiric coverage (ceftriaxone + vancomycin + sometimes ampicillin or steroids) is started immediately and refined when cultures return.

Dura, arachnoid, pia, CSF. Casey's LP is the meninges and the subarachnoid space delivering a diagnosis in a tube.